

Indian Institute of Technology Madras
Department of Mathematics
MA2031-Linear Algebra for Engineers
End Semester Examination

Time : 2 PM-5 PM

Date : May 01, 2023

Maximum Marks : 60

INSTRUCTIONS :

- This question paper contains two parts. Part A and Part B.
- Question numbers 1 to 6 are in Part A. Answer them in the answer script marked A.
- Question numbers 7 to 12 are in Part B. Answer them in the answer script marked B.
- Please write your name, roll number and teacher's name on the top page of both the answer scripts A and B.

Part A - Answer all the six questions - 26 Marks

1. Let $W = \left\{ A \in \mathbb{R}^{2 \times 2} \mid A \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A \right\}.$

(a) Show that W is a subspace of $\mathbb{R}^{2 \times 2}$. [1]

(b) Find a proper subset $S \subset W$ such that $W = \text{Span}\{S\}$. [2]

(c) Find a basis for W . [1]

(d) Determine the dimension of W . [1]

2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear map defined by $T(x, y, z) = (x + 2y + 3z, 4x + 5y + 6z)$. Determine the adjoint T^* . [3]

3. Consider $\mathbb{C}^3$ with the usual inner product. Define $T : \mathbb{C}^3 \rightarrow \mathbb{C}$ by $T(x, y, z) = 3x + iy + 5z$. Find a vector $w \in \mathbb{C}^3$ such that $T(v) = \langle v, w \rangle$ for each $v \in \mathbb{C}^3$. [3]

4. Let $\mathbb{R}_2([0, 1])$ be the vector space of all real polynomial functions of degree at most 2 defined on $[0, 1]$ with inner product given by $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$. Let $T : \mathbb{R}_2([0, 1]) \rightarrow \mathbb{R}_2([0, 1])$ be a linear map defined by $T(a_0 + a_1t + a_2t^2) = a_1t$, and $B = \{1, t, t^2\}$.

- (a) Compute $\langle T(1), t \rangle$. Use this to conclude that $T \neq T^*$. [1]
 (b) Show that $M_{BB}(T) = (M_{BB}(T))^*$. [2]
 (c) Although $M_{BB}(T)$ is hermitian, $T \neq T^*$. Explain why is this not a contradiction. [2]

5. Consider $W = \{(x, y) \in \mathbb{R}^2 \mid 4x + 3y = 0\}$ and let $x = (2, 4) \in \mathbb{R}^2$.

- (a) Find the best approximation to x from W . [3]
 (b) Write $x = a + b$, where $a \in W$ and $b \in \mathbb{R}^2$ such that $\langle a, b \rangle = 0$. [2]

6. Justify why the following matrices are diagonalizable or not.

(a)
$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$
 [2]

(b)
$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$
 [2]

(c) A^*A , for any $A \in \mathbb{C}^{m \times n}$. [1]

Part B-Answer all the six questions - 34 Marks

7. Consider $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & 2 & -1 \\ 2 & 2 & 0 \end{bmatrix}$.

- (a) Determine the spectrum $\sigma(A)$. [2]
 (b) Determine all the Eigenspaces. [4]

8. Let $\mathbb{R}_2([-1, 1])$ be the vector space of all real polynomial functions of degree at most 2 defined on $[-1, 1]$.

(a) Convert the basis $\{1, t, t^2\}$ to an orthonormal basis with respect to the inner product $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt$. [3]

(b) Using part (a), find a polynomial $q \in \mathbb{R}_2([-1, 1])$ such that $\int_{-1}^1 p(t)q(t)dt = p(0)$ for all $p \in \mathbb{R}_2([-1, 1])$. [2]

9. Find an upper triangular matrix which is similar to the matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & -2 & 2 \\ 2 & -3 & 2 \end{bmatrix}$. [6]

✓ 10. Obtain the QR decomposition of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. [6]

✓ 11. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear map defined by

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

(a) Determine the range and null space of T . [4]

(b) Is T an isomorphism? Justify the answer. [2]

✓ 12. Let A be a 3×3 real matrix satisfying $A^2 = 2A + 3I$, where I is the 3×3 identity matrix, and the determinant of A is -9 .

(a) Find the minimal polynomial of the matrix A . [3]

(b) Find the trace of the matrix A . [2]